

Mean-Field Variational Inference for Credal Networks

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Abstract

We describe a mean-field variational inference algorithm for credal networks that computes bounds on marginal probabilities by exploring combinations of extreme points from the local credal sets. For each selected combination of extreme points, the credal network reduces to a standard Bayesian network on which classical mean-field coordinate ascent is applied. By tracking the minimum and maximum of the variational distribution across all explored combinations, the algorithm produces probability intervals for the query variable. This approach complements the interval belief propagation method by providing an alternative view of the credal inference problem through the lens of variational optimization.

1 Preliminaries

1.1 Mean-Field Variational Inference

Given a probability distribution $P(\mathbf{X})$ that factorizes according to a Bayesian network, the **mean-field** approximation seeks a factored distribution $q(\mathbf{X}) = \prod_{i=1}^n q_i(X_i)$ that minimizes the KL divergence:

$$\text{KL}(q\|P) = \sum_{\mathbf{x}} q(\mathbf{x}) \log \frac{q(\mathbf{x})}{P(\mathbf{x})}. \quad (1)$$

The optimal update for each factor $q_i(X_i)$ is:

$$\log q_i^*(X_i) = \mathbb{E}_{q_{\setminus i}}[\log P(\mathbf{X})] + \text{const}, \quad (2)$$

where $q_{\setminus i}$ denotes the product of all factors except q_i .

For a Bayesian network $P(\mathbf{X}) = \prod_i P(X_i \mid \text{Pa}(X_i))$, Equation (??) decomposes into contributions from the local CPTs involving X_i (either as child or parent):

$$\log q_i^*(X_i = x) = \sum_{f: X_i \in \text{scope}(f)} \mathbb{E}_{q_{\setminus i}}[\log P_f], \quad (3)$$

where P_f denotes the CPT of factor f evaluated at the relevant configuration.

1.2 Extension to Credal Networks

In a credal network, each local CPT $P(X_i \mid \text{Pa}(X_i))$ is not a single distribution but a member of a credal set $K(X_i \mid \text{Pa}(X_i))$. The strong extension (Equation (??) below) is the convex hull of all joint distributions formed by selecting one extreme point per local credal set:

$$\text{co } K(\mathbf{X}) = \text{CH} \left\{ \prod_i P(X_i \mid \pi_i) : P(X_i \mid \pi_i) \in \text{ext } K(X_i \mid \pi_i) \right\}. \quad (4)$$

Our approach exploits this structure: for each combination of extreme points (one per local credal set per parent configuration), we obtain a standard Bayesian network. Running mean-field variational inference on this BN produces a specific $q(\mathbf{X})$. By tracking the range of $q(Z)$ across all explored combinations, we obtain bounds on $P(Z \mid \mathbf{Y} = \mathbf{y})$.

2 Algorithm

Algorithm 1 Mean-Field Variational Inference for Credal Networks

Require: Factor graph with extreme points, query Z , evidence $\mathbf{Y} = \mathbf{y}$, max iterations T , threshold τ , max vertex combinations M

Ensure: Bounds $[\underline{P}(Z=z), \overline{P}(Z=z)]$ for each z

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1:  $P(z) \leftarrow 1, \overline{P}(z) \leftarrow 0$  for each  $z$ 
2: Enumerate (or sample) vertex combinations  $\mathcal{C}$  ▷ Up to  $M$  combinations

3: for each vertex combination  $\mathbf{vc} \in \mathcal{C}$  do ▷ FIX EXTREME POINTS  $\rightarrow$  PRECISE BN
4:   Initialize  $q_i(X_i) \leftarrow \text{Uniform}(D_i)$  for all non-evidence  $X_i$ 
5:   Set  $q_j(Y_j) \leftarrow \delta_{y_j}$  for each evidence variable  $Y_j$ 

6:   for  $t = 1, \dots, T$  do ▷ COORDINATE ASCENT
7:      $\delta \leftarrow 0$ 
8:     for each non-evidence variable  $X_i$  do
9:        $\log \tilde{q}_i(x) \leftarrow 0$  for each state  $x$ 
10:      for each factor  $f$  with  $X_i \in \text{scope}(f)$  do
11:        Compute contribution to  $\log \tilde{q}_i$  using  $\mathbf{vc}$  and current  $q_{\setminus i}$  ▷ See Section ??
12:      end for
13:       $q_i^{\text{new}} \leftarrow \text{normalize}(\exp(\log \tilde{q}_i))$ 
14:       $\delta \leftarrow \max(\delta, \|q_i^{\text{new}} - q_i\|_\infty)$ 
15:       $q_i \leftarrow q_i^{\text{new}}$ 
16:    end for
17:    if  $\delta < \tau$  then break
18:    end if
19:  end for

20:  for each state  $z$  do ▷ TRACK BOUNDS
21:     $\underline{P}(z) \leftarrow \min(\underline{P}(z), q_Z(z))$ 
22:     $\overline{P}(z) \leftarrow \max(\overline{P}(z), q_Z(z))$ 
23:  end for
24: end for
25: return  $[\underline{P}(z), \overline{P}(z)]$  for each  $z$ 

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2.1 Coordinate Ascent Update

For a given vertex combination $\mathbf{vc}$, let $P_{\mathbf{vc}}(X_i \mid \pi)$ denote the selected extreme point for node X_i and parent config π . The contribution of factor f (associated with node X_j) to $\log \tilde{q}_i$ is:

Case 1: $X_i = X_j$ (factor's CPT is for X_i itself).

$$\log \tilde{q}_i(x) += \sum_{\pi} \left(\prod_{w \in \text{Pa}(X_i)} q_w(\pi_w) \right) \cdot \log P_{\mathbf{vc}}(X_i=x \mid \pi). \quad (5)$$

Case 2: $X_i \in \text{Pa}(X_j)$ (X_i is a parent in this factor).

$$\log \tilde{q}_i(x) += \sum_{\pi: \pi_i=x} \left(\prod_{w \in \text{Pa}(X_j) \setminus \{X_i\}} q_w(\pi_w) \right) \cdot \sum_{x_j} q_j(x_j) \log P_{\mathbf{vc}}(X_j=x_j \mid \pi). \quad (6)$$

In both cases, the contribution is the expected log-CPT under the current variational distribution over the other variables.

3 Vertex Combination Sampling

The total number of vertex combinations is the product of the number of extreme points across all local credal sets and parent configurations. For the alarm network, this is $2 \times 2 \times 2^4 \times 4 \times 11 = 2816$. For larger networks, the number can be enormous.

To handle this, the algorithm samples up to M combinations uniformly at random (default $M = 500$). This provides a stochastic exploration of the credal set space. Increasing M produces tighter bounds at the cost of computation time.

Exhaustive vs. sampling. When the total number of combinations is $\leq M$, all combinations are explored exhaustively. Otherwise, M combinations are sampled with a fixed random seed for reproducibility.

4 Running Example

Example 1 (Alarm Network). *Using the alarm network with variables B , E , A , and CD .*

Query: $P(A=1 \mid B=0, E=0)$. **Vertex combination 1:** Select $P(B) = [0.8, 0.2]$, $P(E) = [0.9, 0.1]$, $P(A \mid B=0, E=0) = [0.956, 0.044]$, and some vertex for $P(CD \mid A)$.

After coordinate ascent:

- $q(A)$: initialized to $[0.5, 0.5]$.
- Update $q(A)$ using the expected log-CPT under $q(B) = \delta_0$, $q(E) = \delta_0$ (evidence): $\log \tilde{q}(A=1) \propto \log P(A=1 \mid B=0, E=0) = \log 0.044$.
- After normalization: $q(A) \approx [0.956, 0.044]$.

Vertex combination 2: Select $P(A \mid B=0, E=0) = [1.0, 0.0]$. After coordinate ascent: $q(A) \approx [1.0, 0.0]$ ($A=1$ gets probability near 0).

Tracking bounds:

$$\begin{aligned} \underline{P}(A=1) &= \min(0.044, 0.0, \dots) \approx 0.0, \\ \overline{P}(A=1) &= \max(0.044, 0.0, \dots) \approx 0.044. \end{aligned}$$

Over 500 sampled vertex combinations, the tracked bounds were:

<i>Method</i>	$\underline{P}(A=1 \mid B=0, E=0)$	$\overline{P}(A=1 \mid B=0, E=0)$
<i>Exact VE</i>	<i>0.0000</i>	<i>0.0444</i>
<i>Variational (500 combos)</i>	<i>0.0356</i>	<i>1.0000</i>

The variational bounds are wider than the exact VE bounds. This reflects the fact that mean-field variational inference introduces additional approximation through the factored q assumption, and the sampling of vertex combinations may not cover the full extremal range. The bounds improve with more combinations and on larger networks where VE is intractable.

5 Comparison with Other Methods

Property	Exact VE	ε -VE	Interval BP	Variational
Exact	Yes	No	No	No
Bound type	Tight	Outer	Outer	Inner/Outer
Multi-valued	Yes	Yes	Yes	Yes
Scalability	Low	Medium	High	Medium

6 Complexity

For each vertex combination, one round of mean-field coordinate ascent takes $O(n \cdot k \cdot S_{\max} \cdot k^{S_{\max}})$ time, where n is the number of variables, k is the maximum domain size, and $S_{\max}$ is the maximum factor scope size. With T iterations and M sampled combinations, the total complexity is $O(M \cdot T \cdot n \cdot k^{S_{\max}+1})$.

7 References

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